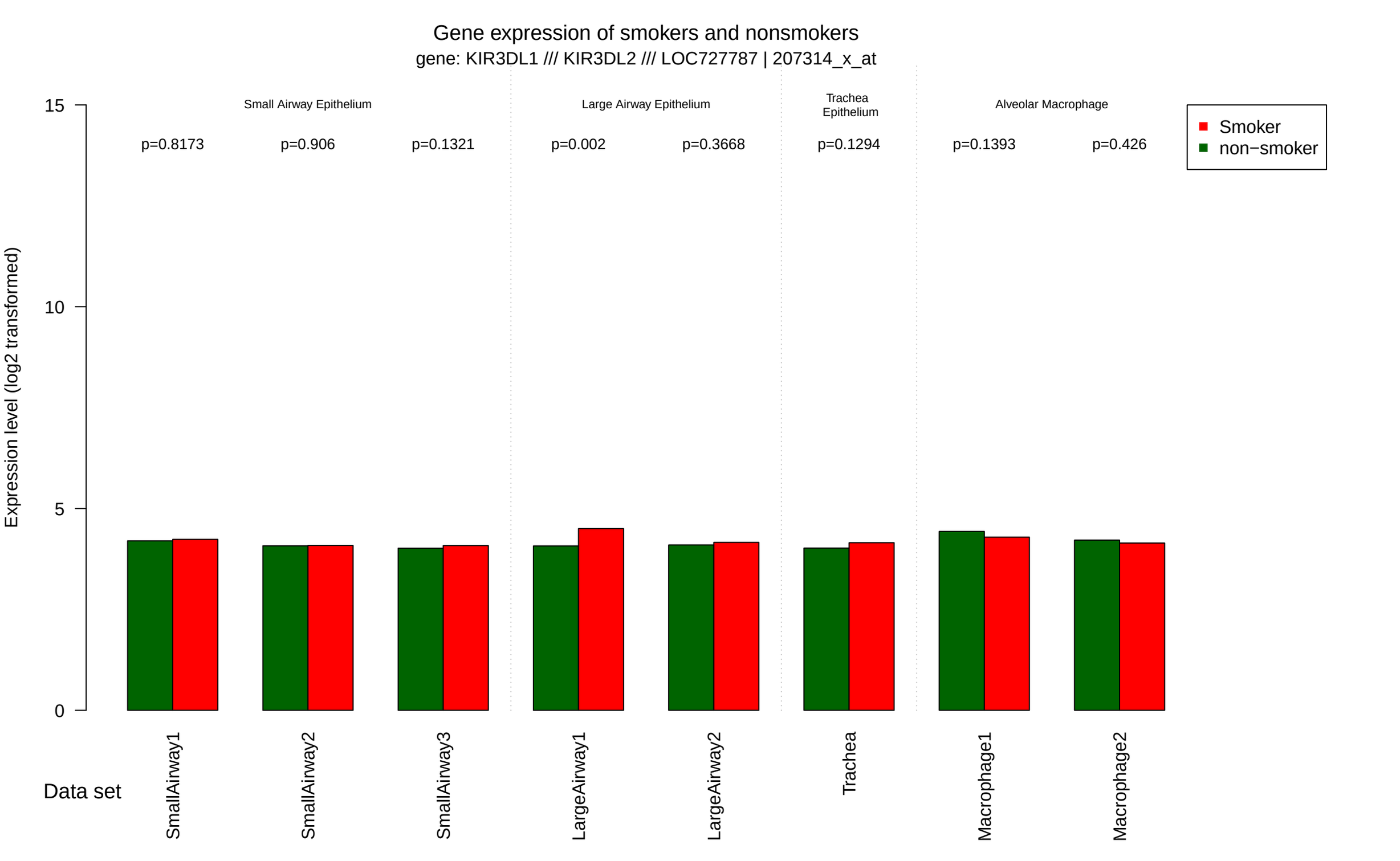
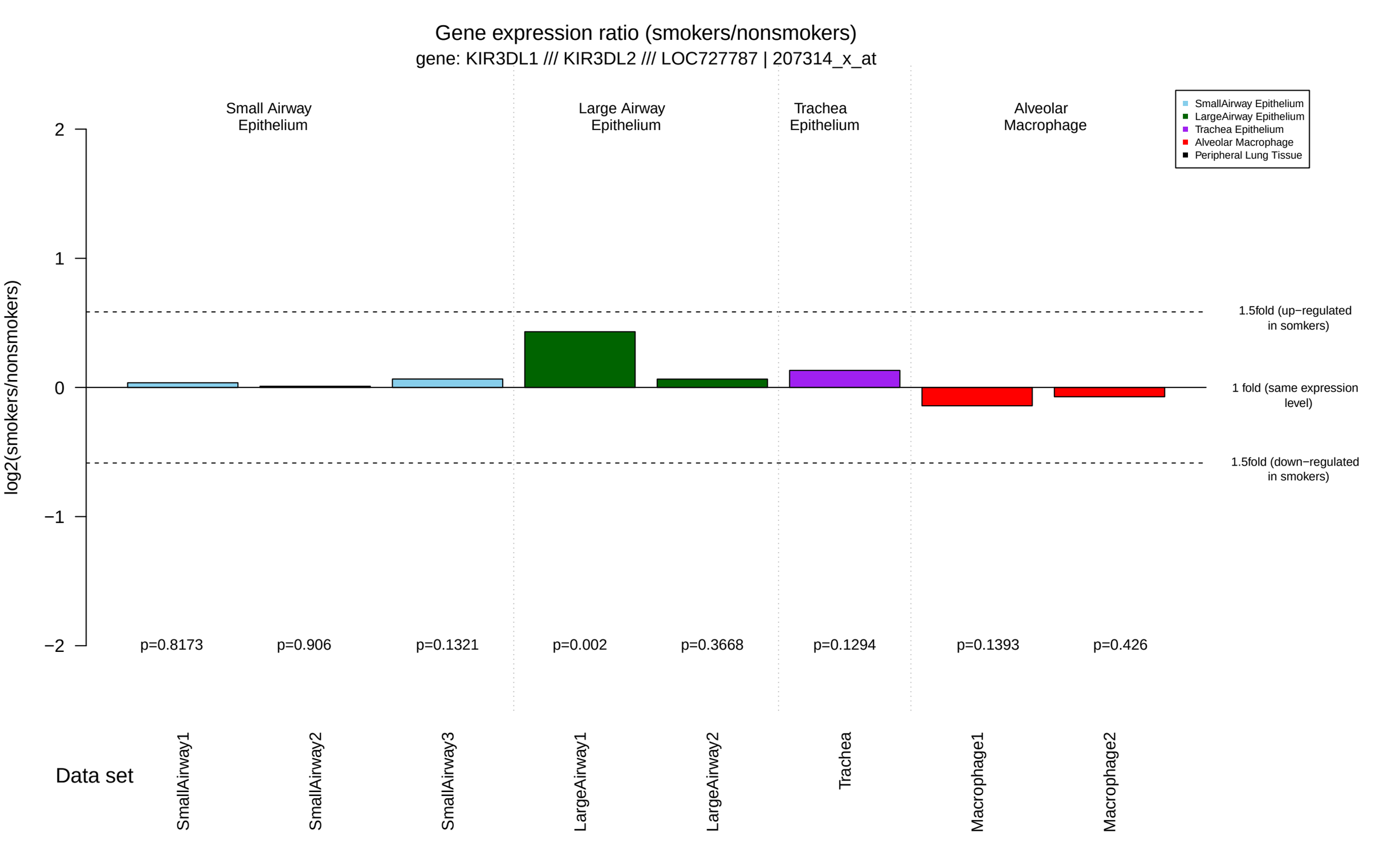
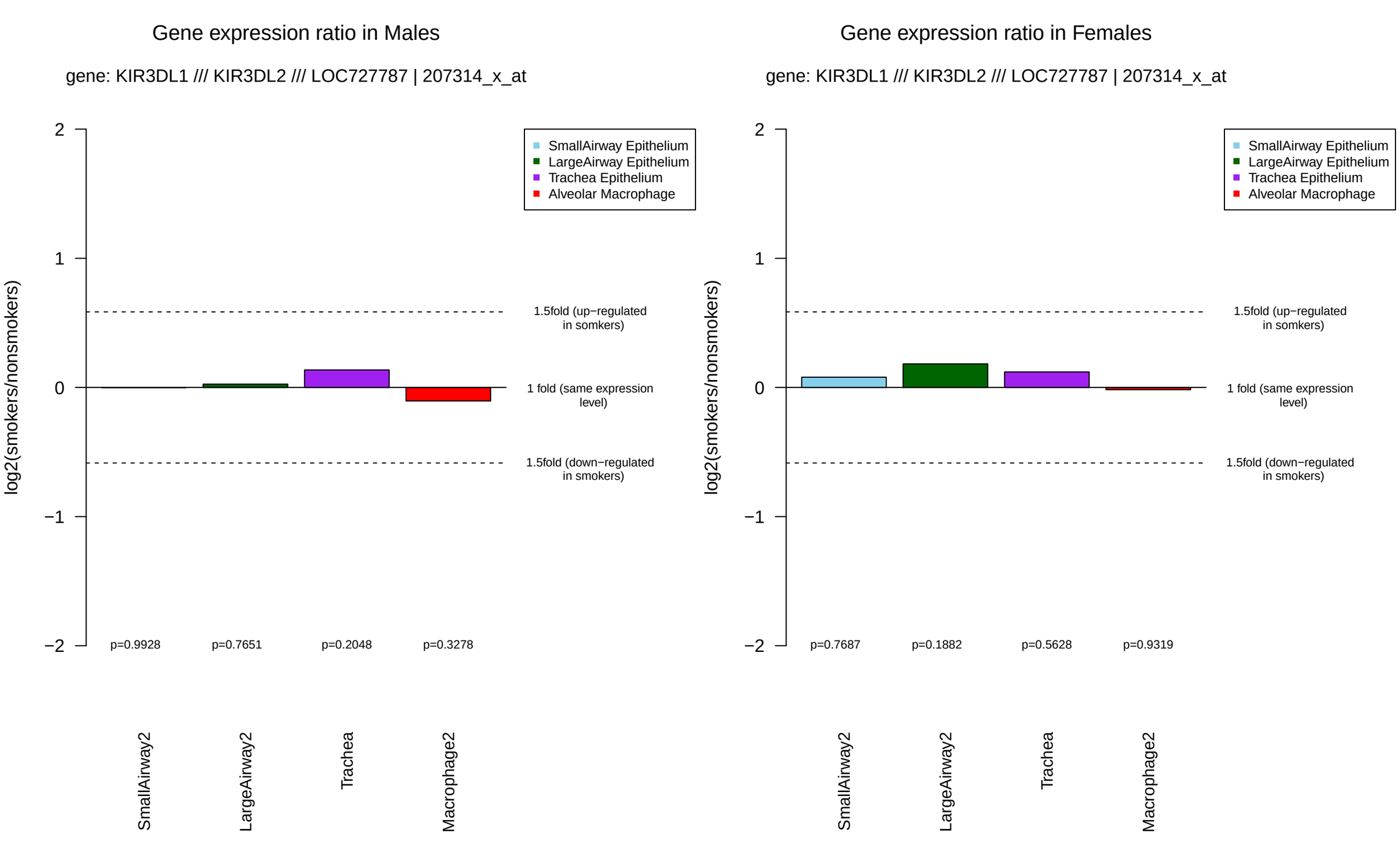
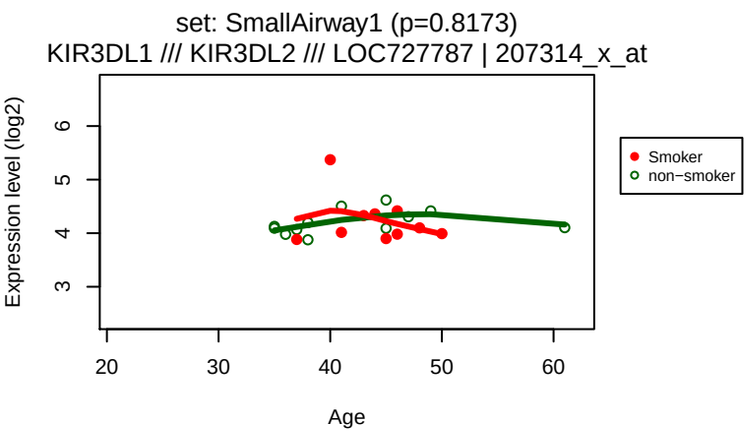
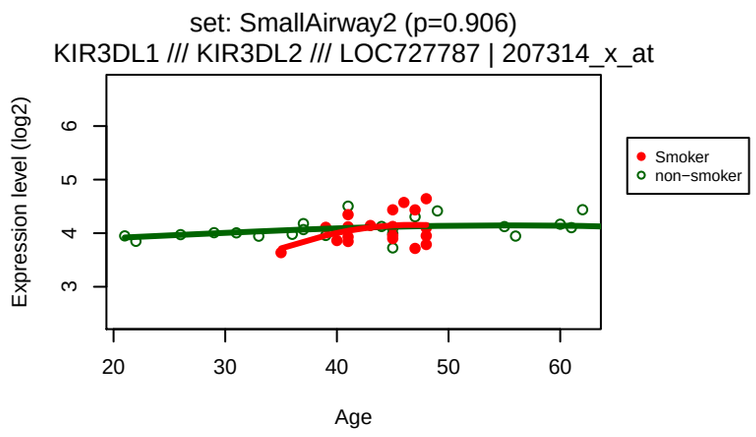
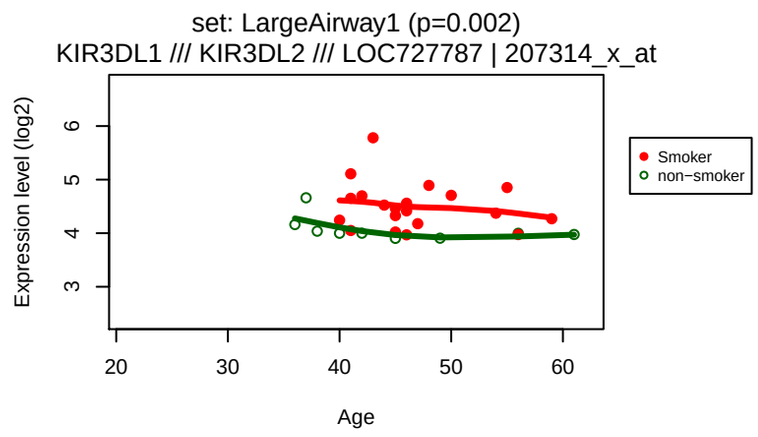

Gene expression of smokers and nonsmokers
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8173
p=0.906
p=0.1321
p=0.002
p=0.3668
p=0.1294
p=0.1393
p=0.426
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

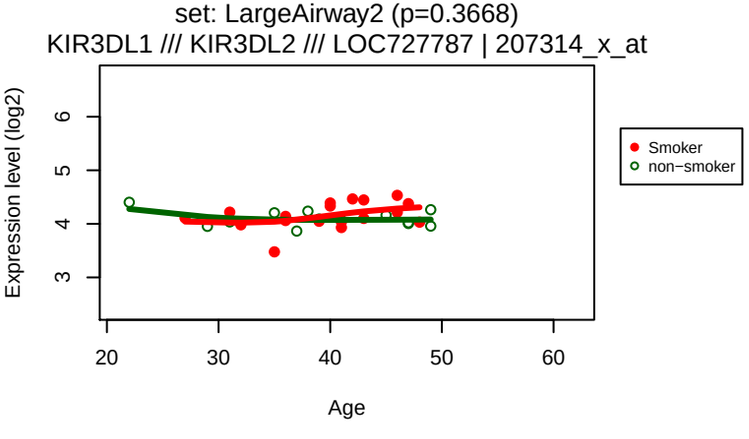

Gene expression ratio (smokers/nonsmokers)
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8173
p=0.906
p=0.1321
p=0.002
p=0.3668
p=0.1294
p=0.1393
p=0.426
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

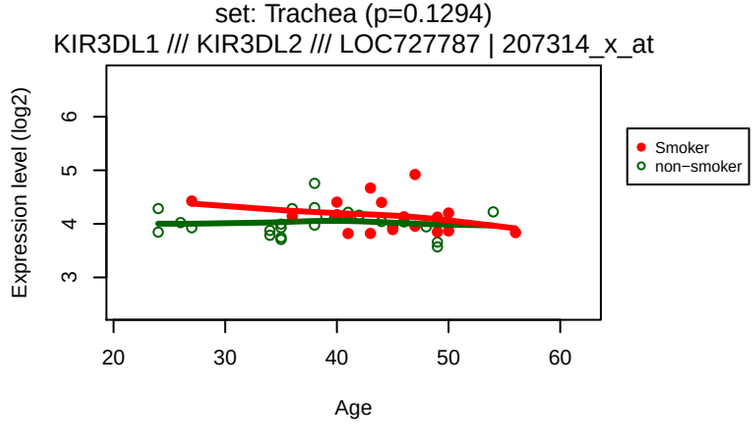

Gene expression ratio in Males
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9928
p=0.7651
p=0.2048
p=0.3278
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7687
p=0.1882
p=0.5628
p=0.9319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

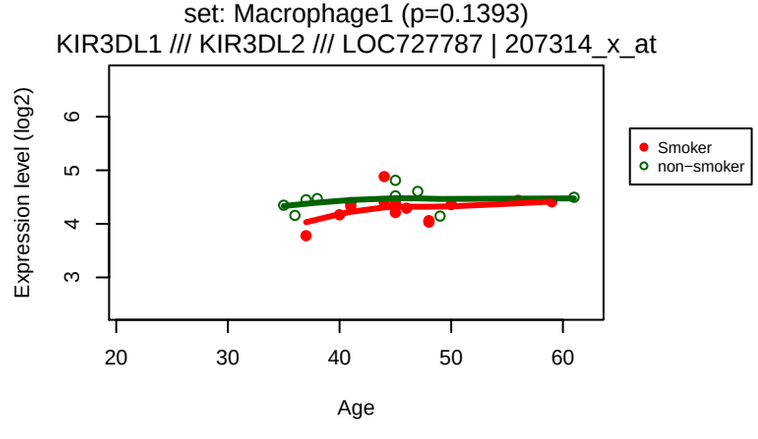

set: SmallAirway1 (p=0.8173)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Expression level (log2)
Age
Smoker
non−smoker

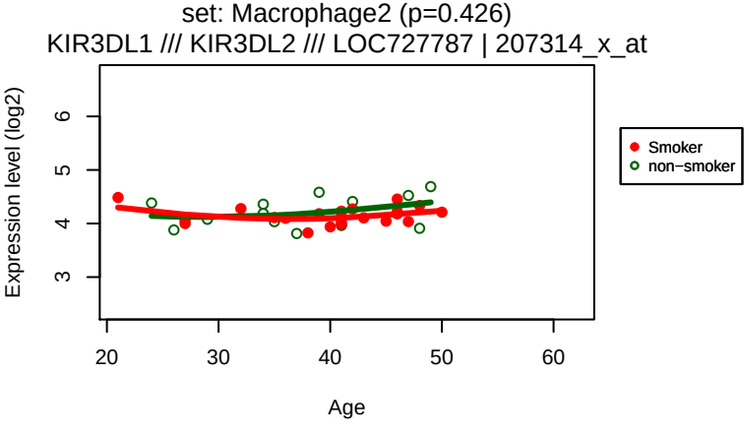

set: SmallAirway2 (p=0.906)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Expression level (log2)
Age
Smoker
non−smoker

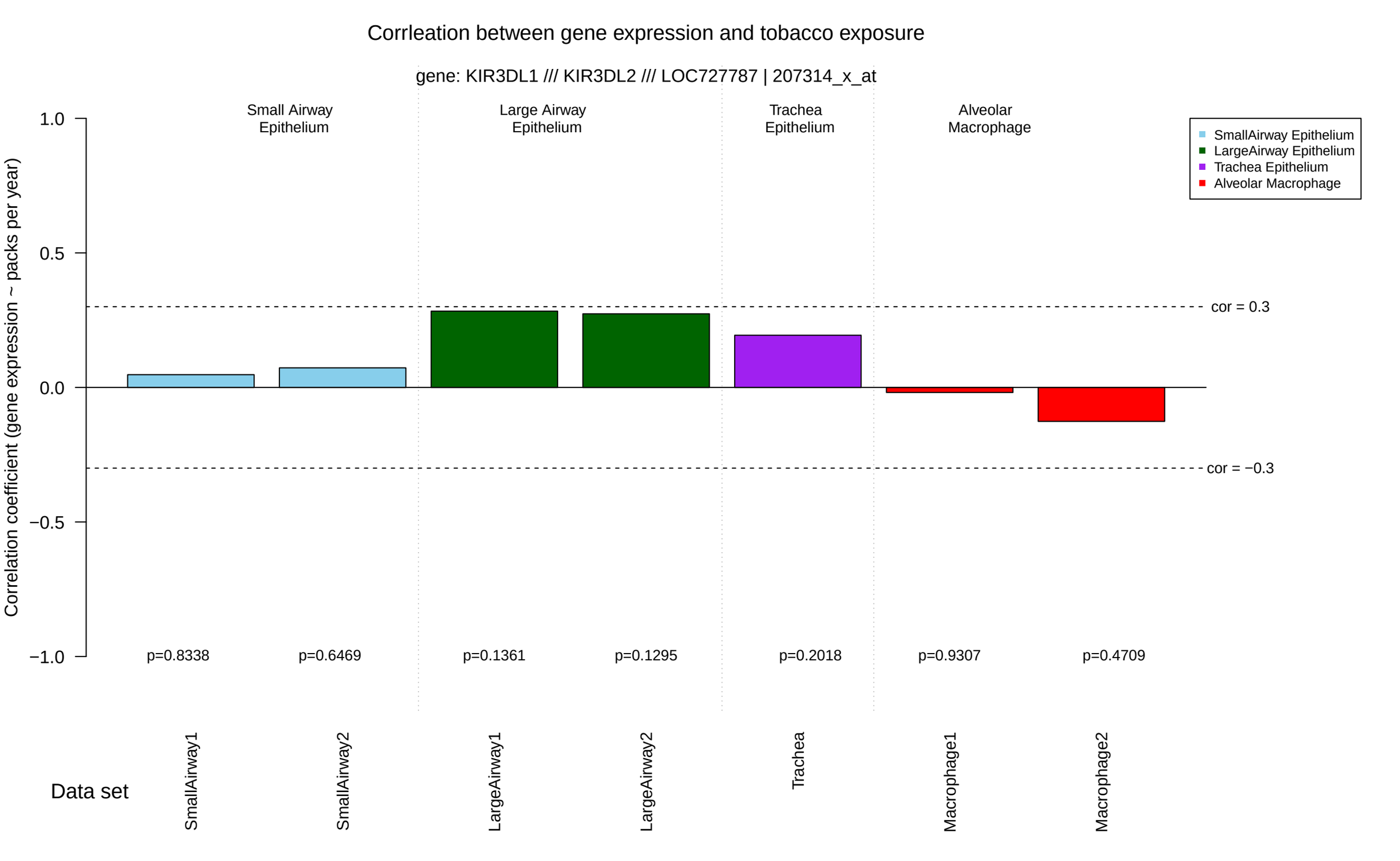

set: LargeAirway1 (p=0.002)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3668)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1294)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1393)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.426)
KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIR3DL1 /// KIR3DL2 /// LOC727787 | 207314_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8338
p=0.6469
p=0.1361
p=0.1295
p=0.2018
p=0.9307
p=0.4709
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2